

ESL MEASUREMENT AND SWITCHING TECHNOLOGY ALTERNATIVES

A PROJECT REPORT
SUBMITTED IN FULFILLMENT OF
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ABSTRACT

The following report covers 2 projects: ESL Measurement & Switching Technologies and their progress during my 6 week summer internship.

Project 1 explains how the 'low voltage surge discharge method' can be used to get accurate readings for 2 terminal capacitors and a proof of concept for how this method can be used for 4 terminal capacitors as well. The aim is to provide a reliable alternative to the Keysight Impedance Analyzer for lab testing of the ESL of film capacitors. Project 2 provides a literature review of available switching technologies available in the industry and shows why SiC MOSFET can be used as an alternative for mercury switches in capacitor charge-discharge circuit.

Keywords: ESL Measurement, SiC MOSFET.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

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Abbreviations

ABB	Auto-Balanced Bridge
ANOVA	Analysis of Variance
AV	Appraiser Variance (Reproducibility)
CPW	Coplanar Waveguide
DAQ	Data Acquisition System
DC	Direct Current
DUT	Device Under Test
EMI	Electromagnetic Interference
ESL	Equivalent Series Inductance
ESR	Equivalent Series Resistance
EV	Equipment Variance (Repeatability)
GaN HEMT	Gallium Nitride High-Electron-Mobility Transistor
IR	Insulation Resistance
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
MSE	Mean Squared Error
MWR	Mercury-Wetted Relay
OTA	Operational Transimpedance Amplifier
RoHS	Restriction of Hazardous Substances
SCOMET	Special Chemicals, Organisms, Materials, Equipment and Technologies
SCR	Silicon Controlled Rectifier (Thyristor)
SiC	Silicon Carbide
SRF	Self-Resonant Frequency
TDR	Time Domain Reflectometry
TDT	Time Domain Transmission
WBG	Wide-Bandgap Semiconductor

Symbols & Nomenclature

Z_x	Complex impedance vector of the DUT
R_s / R	Apparent series resistance / Equivalent series resistance
X_s	Apparent series reactance
ω	Angular frequency
C_s	Apparent series capacitance
C	True frequency-independent capacitance
V_x	Applied AC voltage stimulus
I_x	AC current flowing through the DUT
V_r	Balance voltage across the active range resistor
R_r	Range feedback resistor value
θ	Impedance phase angle
σ^2	Variance parameters (Total, System, Part, Repeatability, Reproducibility)
$i(t)$	Transient discharge current as a function of time
$L / L_{\text{loop}} / L_{\text{fixture}}$	Inductance (True ESL / Loop inductance / Fixture parasitic)
α	Neper frequency (damping factor)
ω_0	Undamped natural resonant frequency
ω_d	Damped natural resonant frequency
$s_{1,2}$	Characteristic roots of the governing series RLC equation
V_0 / V_{op}	Pre-charge target capacitor voltage / Operating voltage
T_d	Ringing waveform damped oscillation period
δ	Logarithmic decrement of peak transient currents
I_1, I_2	Current values at consecutive positive peaks
$V_C / V_1, V_2$	Voltage across capacitor core / Transient peak voltage markers
ΔV	Voltage step drop at pulse initiation ($t \rightarrow 0^+$)
di/dt	Current derivative (ramp rate)
$\Gamma(s)$	Laplace-domain reflection coefficient
Z_0	Coaxial path characteristic impedance (typically 50Ω)
τ	Reflection exponential decay time constant
$\rho(t)$	Time-domain reflection coefficient function
σ	Statistical standard deviation

CHAPTER 1

INTRODUCTION

1.1 General

In high-power electronic systems, characterizing the Equivalent Series Inductance (ESL) of DC-link and snubber film capacitors is critical for managing voltage overshoot during rapid semiconductor switching transitions. As modern power systems transition to Wide-Bandgap (WBG) semiconductors, these capacitors must operate with extremely low parasitic values, typically in the single-to-double-digit nanohenry (nH) range. The importance of precision in this regime cannot be overstated; as even a small error (in 10^{-9} range), is often enough to overshadow the internal ESL of the device under test (DUT). Furthermore, SCOMET List by the Directorate General of Foreign Trade has strict limits on inductance of capacitors (10nH or 50nH). This highlights the importance of a reliable ESL measurement method.

The switch element used to initiate the discharge transient is central to achieving this measurement fidelity. The field of pulsed power engineering is undergoing a definitive strategic shift from traditional mechanical switching technologies toward advanced solid-state semiconductor solutions — a transition essential for modern high-precision film capacitor testing. This demand is further reinforced by the urgent requirement for environmental sustainability, specifically the global regulatory phase-out of mercury switches mandated by the Minamata Convention and RoHS compliance.

1.2 Background

Capacitor inductance measurement is inherently complex due to ultra-low target values. Industry standards such as IEC 61071 define self-inductance as a black-box parameter measured under specific terminal conditions. Traditionally, measurements are performed using an Auto-Balanced Bridge (ABB) method with high-precision impedance analyzers (e.g., Keysight E4990A). The Low-Voltage Surge Discharge Method is a time-domain technique that extracts ESL by measuring the underdamped ringing current waveform produced when a pre-charged capacitor discharges

through a low-impedance switch as described in the TDK/EPCOS report.

Historically, the switching element in such discharge setups relied on mercury-wetted relays (MWRs) which currently face environmental restrictions.

1.3 Motivation

This project was motivated by the need to bridge the gap between high-precision time consuming laboratory characterization and high-volume time efficient reliable setup. Existing surge-discharge setups are highly sensitive to topology; requiring constant fixture calibration. A small change can prove fatal, leading to invalid data. The industry is simultaneously facing a mandatory transition away from MWRs due to Minamata Convention toxicity regulations, creating an urgent need to validate solid-state switching alternatives. Adopting SiC MOSFET technology directly addresses the dual problem of environmental toxicity and measurement accuracy, enabling reduction in switching jitter and system footprint.

1.4 Objectives of the Project

The specific objectives of this project are:

1. Validate and standardize the low-voltage surge-discharge method for film capacitor ESL measurement to ensure a repeatable, traceable workflow for high volume R&D testing.
2. Design a high-precision metrology fixture, such as a laminated H-bridge busbar for increasing accuracy of 4 terminal capacitors.
3. Evaluate, characterize and provide a proof of concept to integrate Silicon Carbide (SiC) MOSFET switching technology .
4. Integrate project 1 & project 2 as a unified ESL Measuring setup.

1.5 Organization of the Dissertation

The remainder of this report is structured as follows:

- **Chapter 2** presents a comprehensive literature review of multiple ESL measurement techniques and a comparative analysis of conventional and wide-bandgap switching technologies.
- **Chapter 3** describes the proposed methodology, detailing the mathematical frameworks for transient RLC analysis.
- **Chapter 4** covers the experimental setup, focusing on charge-discharge setup and the mechanical design of the H-bridge fixture

- **Chapter 5** presents results and analysis of tests across variety of capacitors and it's cross-validation.
- **Chapter 6** concludes the report with key findings, practical recommendations and future metrology improvements.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter surveys the existing literature relevant to this project, covering: (1) Laboratory metrology standards for nanohenry-scale inductance characterization, (2) Legacy and high-speed switching technologies for transient capacitor discharge.

2.2 ESL Measurement Methods

2.2.1 Auto-Balanced Bridge (ABB) Method

Theoretical Foundation:

The Auto-Balanced Bridge (ABB) architecture is the primary instrumentation standard for high-precision impedance measurements from low frequencies up to approximately 120MHz. The working mechanism relies on maintaining a virtual ground potential at the low terminal of the Device Under Test (DUT), thereby neutralising the effect of parasitic impedances to ground and isolating the complex impedance vector of the film capacitor.

The complex impedance of a real metallized film capacitor is modelled by the series equivalent circuit:

$$Z_x = R + j\left(\omega L - \frac{1}{\omega C}\right) \quad (2.1)$$

In the ABB configuration, a known AC voltage stimulus V_x is applied to the High-Force (H_{cur}) terminal of the DUT. The resulting current I_x flows through the capacitor to the Low-Force (L_{cur}) terminal. An active Operational Transimpedance Amplifier (OTA) detects any current flow at the L_{cur} node and dynamically adjusts its output voltage V_r through a known range-feedback resistor R_r to force the potential at the L_{cur} terminal to exactly zero volts. The nodal current-balancing

equation at the virtual ground is:

$$I_x + I_r = 0 \implies \frac{V_x}{Z_x} + \frac{V_r}{R_r} = 0 \quad (2.2)$$

Solving for the complex impedance vector Z_x :

$$Z_x = -R_r \cdot \frac{V_x}{V_r} \quad (2.3)$$

Decomposing the voltage ratio into real and imaginary parts, $\frac{V_x}{V_r} = A + jB$, and substituting:

$$Z_x = -R_r(A + jB) = R_s + jX_s \quad (2.4)$$

where the apparent series resistance and reactance are:

$$R_s = -R_r A \quad (2.5)$$

$$X_s = -R_r B \quad (2.6)$$

At a single test frequency ω , a conventional instrument reports the apparent series capacitance C_s based on a simplified two-element capacitive model:

$$C_s = -\frac{1}{\omega X_s} = \frac{1}{\omega R_r B} \quad (2.7)$$

However, near or above the self-resonant frequency (SRF), the inductive reactance counteracts the capacitive reactance, leading to a systematic overestimation of C_s . To isolate the true frequency-independent capacitance C and the nanohenry-range *ESL* (L), measurements must be performed at two distinct frequencies ω_1 and ω_2 within the capacitive regime:

$$X_{s1} = \omega_1 L - \frac{1}{\omega_1 C} \quad (2.8)$$

$$X_{s2} = \omega_2 L - \frac{1}{\omega_2 C} \quad (2.9)$$

Multiplying both equations by their respective angular frequencies:

$$\omega_1 X_{s1} = \omega_1^2 L - \frac{1}{C} \quad (2.10)$$

$$\omega_2 X_{s2} = \omega_2^2 L - \frac{1}{C} \quad (2.11)$$

Subtracting to eliminate the capacitive term yields the true ESL:

$$L = \frac{\omega_1 X_{s1} - \omega_2 X_{s2}}{\omega_1^2 - \omega_2^2} \quad (2.12)$$

Solving for the true capacitance C :

$$C = \frac{\omega_1^2 - \omega_2^2}{\omega_1 \omega_2^2 X_{s1} - \omega_1^2 \omega_2 X_{s2}} \quad (2.13)$$

Analysis:

The phase angle at ω_1 :

$$\theta = \arctan\left(\frac{X_s}{R}\right) = \arctan\left(\frac{-0.5321053}{0.01501}\right) \approx -88.384^\circ \quad (2.14)$$

If this angle tips over -90° due to errors the ABB method reports negative/incorrect values.

Table 2.1: ABB Method – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Phase Detector Resolution	Rectangular	$\pm 0.005^\circ$	$1.25 \times 10^{-10} \text{ H}^\circ^{-1}$	$\pm 0.063 \text{ nH}$
Residual Short Inductance	Normal ($k = 1$)	$\pm 0.150 \text{ nH}$	1.00	$\pm 0.150 \text{ nH}$
Range Resistor Drift	Triangular	$\pm 10 \text{ ppm}/^\circ\text{C}$	$4.20 \times 10^{-11} \text{ H/ppm}$	$\pm 0.042 \text{ nH}$
Guard-Loop Leakage	Rectangular	$\pm 0.05 \mu\text{A}$	$1.80 \times 10^{-4} \text{ H A}^{-1}$	$\pm 0.009 \text{ nH}$
Combined U_c	Normal ($k = 1$)	—	—	$\pm 0.168 \text{ nH}$

Formal Gauge R&R Framework:

The measurement system variance is modelled using ANOVA decomposition:

$$\sigma_{\text{total}}^2 = \sigma_{\text{part}}^2 + \sigma_{\text{system}}^2 = \sigma_{\text{part}}^2 + \sigma_{\text{repeatability}}^2 + \sigma_{\text{reproducibility}}^2 \quad (2.15)$$

Repeatability (Equipment Variance, EV):

$$EV = 5.15 \cdot \sqrt{MSE} \quad (2.16)$$

Reproducibility (Appraiser Variance, AV):

$$AV = \sqrt{(5.15 \cdot \sigma_{\text{operator}})^2 - \frac{EV^2}{n \cdot r}} \quad (2.17)$$

2.2.2 Voltage Surge Discharge Method

Theoretical foundation:

The Voltage Surge Discharge Method is an active time-domain technique used to isolate *ESL* and *ESR* from the underdamped ringing current produced when a pre-charged capacitor discharges through a low-impedance switch. The discharge loop is modelled as a series *RLC* circuit governed by:

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0 \quad (2.18)$$

Dividing by L :

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{LC} i(t) = 0 \quad (2.19)$$

The characteristic roots are:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \quad (2.20)$$

where:

$$\alpha = \frac{R}{2L} \quad (\text{Neper frequency}) \quad (2.21)$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (\text{undamped resonant frequency}) \quad (2.22)$$

For the underdamped condition ($\alpha < \omega_0$), the damped resonant frequency is:

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} \quad (2.23)$$

With initial conditions $q(0) = CV_0$ and $i(0) = 0$:

$$i(t) = \frac{V_0}{\omega_d L} e^{-\alpha t} \sin(\omega_d t) \quad (2.24)$$

The damped period is measured from consecutive zero-crossings:

$$T_d = \frac{2\pi}{\omega_d} \implies \omega_d = \frac{2\pi}{T_d} \quad (2.25)$$

The logarithmic decrement from two consecutive positive peaks I_1, I_2 :

$$\delta = \ln\left(\frac{I_1}{I_2}\right) = \alpha T_d \implies \alpha = \frac{\delta}{T_d} \quad (2.26)$$

Then:

$$\omega_0 = \sqrt{\omega_d^2 + \alpha^2} \quad (2.27)$$

$$L_{\text{loop}} = \frac{1}{\omega_0^2 C} \quad (2.28)$$

$$L_C = \frac{V_1 L_{\text{loop}}}{V_C} \quad (2.29)$$

Analysis:

At very low values of ESL, $V_1 \ll V_C$ making it susceptible to setup noise and measurement errors

Table 2.2: Voltage Surge Discharge – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Digitizer Sampling Jitter	Rectangular	$\pm 500 \text{ ps}$	$3.16 \times 10^{-2} \text{ nH ns}^{-1}$	$\pm 0.016 \text{ nH}$
Switch Dynamic Resistance	Normal ($k = 2$)	$\pm 0.80 \text{ m}\Omega$	$1.50 \times 10^{-9} \text{ H } \Omega^{-1}$	$\pm 0.120 \text{ nH}$
Dummy Block Calibration Error	Triangular	$\pm 0.05 \text{ nH}$	1.00	$\pm 0.035 \text{ nH}$
Coaxial Shunt Thermal Drift	Normal ($k = 1$)	$\pm 0.2\%$	$4.18 \times 10^{-2} \text{ nH } \%^{-1}$	$\pm 0.084 \text{ nH}$
Combined U_c	Normal ($k = 1$)	—	—	$\pm 0.151 \text{ nH}$

2.2.3 Constant Current Pulse(di/dt) Method

Theoretical foundation:

The terminal voltage of a capacitor under dynamic current is governed by:

$$V(t) = L \frac{di}{dt} + i \cdot R_{ESR} + \frac{1}{C} \int i dt \quad (2.30)$$

Term Analysis at $t \rightarrow 0^+$

1. Capacitance Term $\left(\frac{1}{C} \int i dt\right)$: Pulse width $< 2 \mu\text{s} \Rightarrow$ negligible charge accumulation $\Rightarrow \approx 0$
2. ESR Term $(i \cdot R_{ESR})$: At ramp onset, $i \approx 0 \Rightarrow$ negligible
3. ESL Term $\left(L \frac{di}{dt}\right)$: $\frac{di}{dt}$ is massive $\Rightarrow$ **dominates**

Equation simplifies to:

$$\boxed{V(t)|_{t \rightarrow 0^+} \approx L_{ESL} \frac{di}{dt}} \quad (2.31)$$

Waveform Behavior:

For a linear current ramp:

$$i(t) = t \frac{di(t)}{dt} \quad (2.32)$$

The voltage response exhibits:

- An instantaneous vertical jump ΔV at $t = 0^+$ (ESL-dominated)
- A slower linear slope for $t > 0$ (capacitor charging)
- By measuring the immediate vertical jump (ΔV) and knowing the pre-set di/dt , the system calculates the ESL in real time

ESL Extraction

$$L_{ESL} = \frac{\Delta V}{\left(\frac{di}{dt}\right)} \quad (2.33)$$

Analysis:**Table 2.3:** Error Analysis and Gauge R&R for Constant Pulse Method

Category	Parameter	Impact on Measurement Accuracy & Precision
Error Analysis	Transient Signal Detection (ΔV)	Identifying the exact vertical jump (ΔV) at $t \rightarrow 0$ is critical.
	di/dt Linearity	Fluctuations in the current ramp—prevented by precision air-core inductors—introduce direct errors into the $ESL = \Delta V / (di/dt)$ calculation.
	Sampling Resolution	DAQ sampling at < 100 MS/s may miss the peak derivative.
Repeatability (Equipment)	Signal-to-Noise Ratio (SNR)	A 100 A pulse creates a massive, clean, multi-volt jump (ΔV) immune to factory EMI, ensuring consistent results.
	Switching Consistency	SiC MOSFETs provide incredibly sharp and repeatable nanosecond turn-on times.
	Mechanical Rigidity	Rigid pneumatic copper jaws maintain identical geometry for every cycle, keeping fixture inductance ($L_{fixture}$) as a fixed constant.
Reproducibility	Kelvin Contact Jig	Pneumatic actuators ensure sensing probes touch terminals at the absolute closest point, removing risks of manual placement errors.
	System Calibration	Reproducibility across lines is maintained via “golden-master” calibration to subtract the specific fixture constant of each machine.

2.2.4 Time Domain Reflectometry / Transmission (TDR/TDT) Method

Theoretical foundation:

TDR injects an ultra-fast voltage step into a $50\ \Omega$ coaxial path shunt-loaded by the DUT. The Laplace-domain reflection coefficient, assuming $R \ll Z_0$:

$$\Gamma(s) \approx \frac{sL - Z_0}{sL + Z_0} \quad (2.34)$$

For a unit-step input $V_{\text{inc}}(s) = \frac{V_0}{s}$, the inverse Laplace transform yields:

$$v_{\text{meas}}(t) = 2V_0 e^{-\frac{Z_0}{L}t} \cdot u(t) \quad (2.35)$$

The inductance is extracted from the exponential time constant τ :

$$\tau = \frac{L}{Z_0} \implies L = Z_0 \cdot \tau \quad (2.36)$$

For non-ideal (finite rise-time) step excitation, the integration method is used:

$$L = 2Z_0 \int_{t_1}^{t_2} \rho(t) dt \quad (2.37)$$

Analysis:

Table 2.4: TDR Method – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Oscilloscope Timing Jitter	Normal ($k = 1$)	$\pm 500\text{ fs}$	$1.20 \times 10^{-1} \text{ nH ps}^{-1}$	$\pm 0.060 \text{ nH}$
Step Amplitude Fluctuations	Normal ($k = 2$)	$\pm 0.50\%$	$4.20 \times 10^{-2} \text{ nH } \%^{-1}$	$\pm 0.011 \text{ nH}$
CPW Z_0 Machining Tolerance	Rectangular	$\pm 0.25\ \Omega$	$4.20 \times 10^{-10} \text{ H } \Omega^{-1}$	$\pm 0.105 \text{ nH}$
Time Gating Window Drift	Triangular	$\pm 2.0\text{ ps}$	$5.00 \times 10^{-2} \text{ nH ps}^{-1}$	$\pm 0.100 \text{ nH}$
Combined U_c	Normal ($k = 1$)	—	—	$\pm 0.158 \text{ nH}$

Table 2.5: Comparison Matrix

Method	Error distribution	$L_{fixture}$	ΔL	Rank
ABB	Normal(k=1)	YES	± 0.168 nH	3
TDR/TDT	Normal(k=1)	YES	± 0.158 nH	2
Constant di/dt	Normal(k=1)	YES	± 0.167 nH	4
Proposed	Normal(k=1)	NO	± 0.151 nH	1

2.3 Switching Technologies

2.3.1 Thyristor:

Thyristors can block up to 8.5 kV and handle 120 kA. However, literature identifies them as "switch-on only" devices limited by a low current rise rate (di/dt) of less than 1.5 kA/ μ s and slow transition speeds.

2.3.2 Mercury-Wetted Relay (MWR):

Long prized for "bounce-free" picosecond contact closure, MWRs are now categorized as a "laboratory liability". Mechanical jitter and orientation sensitivity make them inherently unstable for automated production environments.

2.3.3 GaN eHEMT:

GaN offers unmatched package inductance (<1 nH) and sub-2 ns edges. However, GaN is practically limited by a severe voltage bottleneck (typically <600 V).

2.3.4 SiC MOSFET:

Successfully bridges the gap between high voltage (>1.2 kV) and nanosecond transition speeds. Using an optimized gate-boosting technique with GaN driver switches, SiC turn-on times are violently slashed to below 2 ns.

2.3.5 Spark Gap:

While capable of handling extreme voltages (>10 kV), spark gaps suffer from plasma-channel drift and stochastic misfiring. Literature highlights their poor repeatability (Gauge R&R) and the requirement for frequent maintenance due to electrode erosion.

2.3.6 Avalanche Transistor:

These provide fast 300–600 ps edges but are limited to very low voltages (<100 V) and exhibit high device-to-device variability.

Table 2.6: Comparative Technical Analysis of Switching Technologies for Capacitor Discharge

Technology	Transition Speed	Max Voltage	di/dt Capability	Repeatability
Thyristor (SCR)	> 1000 ns	≤ 8.5 kV	< 1.5 kA/ μ s	Poor
Mercury Relay (MWR)	1–5 ns	> 1 kV	High (Jittery)	Poor–Fair
GaN eHEMT	< 2 ns	< 600 V	Ultra-High	Excellent
SiC MOSFET (Boosted)	2–10 ns	> 1.2 kV	38.7 kA/μs	Excellent
Spark Gap	1–10 ns	> 10 kV	High (Drifting)	Poor
Avalanche Transistor	300–600 ps	< 100 V	Moderate	Variable

2.4 Gap Analysis

The literature review reveals the following gaps that this project addresses:

1. **Gap 1:** Most methods need fixture compensation/calibration.
2. **Gap 2:** Switch alternative due to regulatory Phase-out of Mercury Switches.
3. **Gap 3:** Method is (2,4) terminal capacitor inclusive.

CHAPTER 3

METHODOLOGY

3.1 ESL Measurement

A film capacitor can be modeled as follows: 3.1

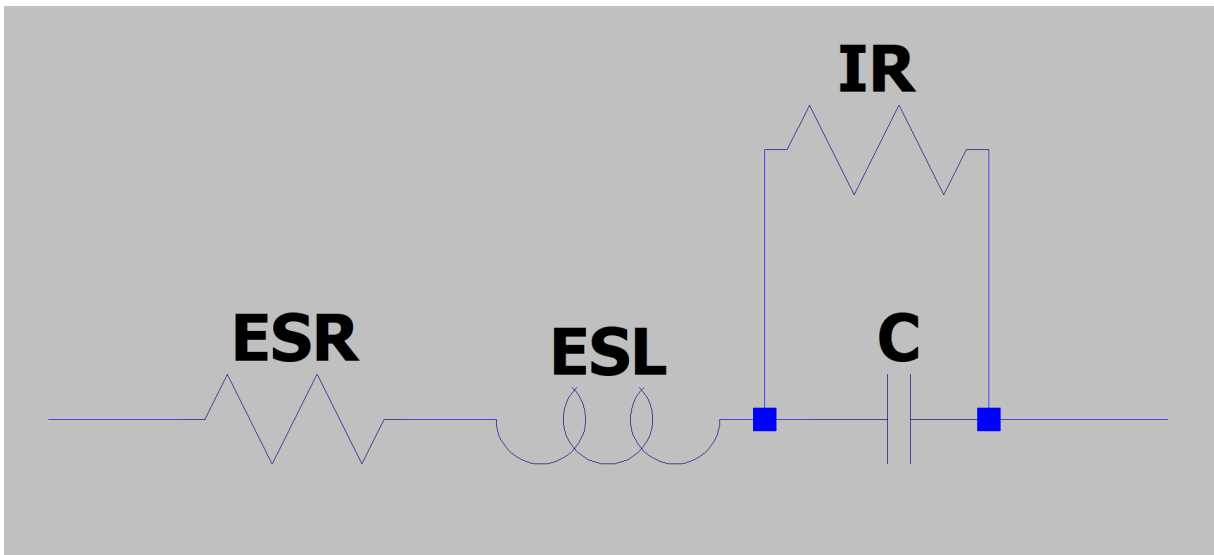


Figure 3.1: [Equivalent Circuit of a Film Capacitor]

where,

ESR stands for effective series resistance

ESL stands for effective series inductance

IR stands for Insulation resistance

Since, IR is very high (>2G Ohm to pass IR test), we treat the capacitor as a series RLC circuit. Thus the voltage drop across the circuit is,

$$V = ESR.i + ESL\frac{di}{dt} + \frac{1}{C} \int i.dt \quad (3.1)$$

Since it is a fast discharge, only inductance reacts instantaneously due to the di/dt term, while other elements follow a little later. Thus, finding ΔV at $t \Rightarrow 0^+$ gives the voltage drop due to inductance only.

This method relies on analysis of discharge waveforms of a capacitor under test. An example waveform is shown below in Figure 3.2.

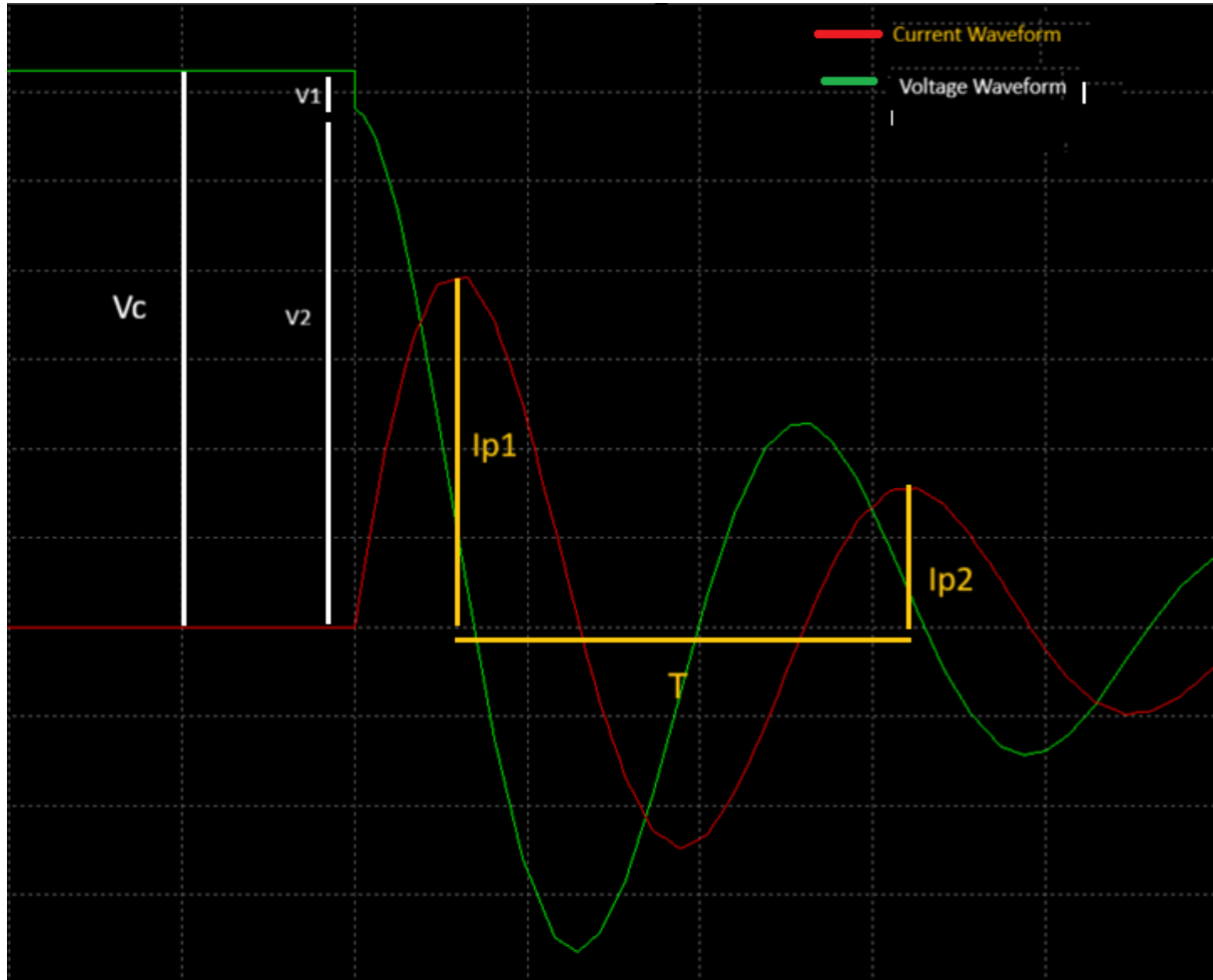


Figure 3.2: [A typical capacitor discharge waveform]

STEP 1: Damping Coefficient

$$e^{\alpha T} = \frac{Ip1}{Ip2} \quad (3.2)$$

$$\alpha = \frac{1}{T} \ln \frac{Ip1}{Ip2} \quad (3.3)$$

also

$$\alpha = \frac{R}{2L} \quad (3.4)$$

STEP 2: Resonance & Natural Frequencies

$$\omega_r = \frac{2\pi}{T} \quad (3.5)$$

$$\omega_o = \sqrt{\omega_r^2 + \alpha^2} \quad (3.6)$$

STEP 3: Loop Inductance

$$\omega_o = \frac{1}{\sqrt{LC}} \quad (3.7)$$

$$L = \frac{1}{\omega_o^2 C} \quad (3.8)$$

STEP 4: Capacitor Inductance

$$\frac{V_1}{V_C} = \frac{L_C}{L} \quad (3.9)$$

$$L_C = L \frac{V_1}{V_C} \quad (3.10)$$

Direct Formula:

$$L_C = \frac{V_1}{V_C} \frac{1}{C} \frac{T^2}{(4\pi^2 + \ln^2 \frac{I_{p1}}{I_{p2}})} \quad (3.11)$$

3.2 SiC Switch

3.2.1 Device Selection and Characterization

The Wolfspeed CAB450M12XM3 is a 1200 V, 450 A SiC MOSFET half-bridge power module. In this work, only the low-side switch of the half-bridge is utilized as the discharge trigger, with the high-side device left unpopulated or held off. Key device parameters governing switching performance are the gate threshold voltage $V_{GS(th)}$, gate charge Q_g , and the internal gate resistance $R_g(int)$. The total gate loop time constant is:

$$\tau_g = (R_g(ext) + R_g(int))C_{iss} \quad (3.12)$$

where C_{iss} is the input capacitance and $R_g(ext)$ is the externally programmable gate resistance. Minimizing τ_g is the primary design lever for achieving fast turn-on and high di/dt at the load. The on-state current rise rate during discharge is governed by the circuit loop inductance L_{loop} and the DC-link voltage V_{DC} :

$$\frac{di}{dt} = \frac{V_{DC}}{L_{loop}} \quad (3.13)$$

3.2.2 Low-Inductance Power Loop Design

The power discharge loop — comprising the capacitor bank, busbar, SiC module, and return path — is designed as a laminated coplanar pair to minimize enclosed loop area. The parasitic inductance of a rectangular loop is estimated as:

$$L_{Loop} = \frac{\mu_o}{\pi} \cdot l \cdot \ln \frac{2h}{w} \quad (3.14)$$

where l is the length of the loop, h is the separation of the conductor, and w is the width of the conductor. Laminating the forward and return conductors with a thin dielectric spacer ($h \Rightarrow \text{minimum}$) is the primary technique to drive L_{loop} extremely low.

3.2.3 CAB450M12XM3 Datasheet

Table 3.1: Key parameters of the chosen SiC switch

Parameter	Symbol	min/typ/max
Drain-Source Voltage	V_{DS}	– / – / 1200 V
Operational Gate-Source Voltage	V_{GS}	-8 / (-4/+15) / +19 V
DC Continuous Drain Current	I_D	– / 438 / 450 A
Pulsed Drain Current	I_{DM}	– / 900 / – A
Power Dissipation	P_D	– / 1670 / – W
Operational Virtual Junction Temperature	T_{VJop}	-40 / +25 / +175 °C
Reverse Recovery Time	t_{RR}	– / 78 / – ns
Reverse Recovery Charge	Q_{RR}	– / 7.2 / – uC
Peak Reverse Recovery Current	I_{RRM}	– / 169 / – A

3.3 Copper H-Bridge

3.3.1 Design Rationale

The central challenge in surge-discharge ESL measurement is ensuring that the fixture’s own parasitic inductance does not dominate the measurement. Standard clip-lead or coaxial fixtures routinely introduce stray inductances of 10–50 nH, which completely mask the single-digit nanohenry ESL of modern DC-link film capacitors. The metrology fixture designed in this work adopts a laminated H-bridge copper busbar geometry, in which the forward and return current paths are forced into extreme proximity. Since stray inductance scales directly with enclosed loop area, minimizing this area is the primary electromagnetic design objective:

$$L_{stray} \propto A_{loop} \quad (3.15)$$

The H-bridge configuration additionally enforces precise Kelvin contact placement and accommodates the DUT, DC supply, and SiC switch within a single low-inductance current loop.

3.3.2 Loop Inductance Minimization (L_{loop})

The total parasitic loop inductance (L_{loop}) of a laminated two-plate assembly carrying equal and opposite currents is defined by the self-inductances (L_1, L_2) of the top and bottom plates and the mutual inductance (M) between them:

$$L_{loop} = L_1 + L_2 - 2M \quad (3.16)$$

While extracting exact closed-form expressions for L and M for finite plates is complex due to fringing fields, the loop inductance can be derived exactly from first principles. Assuming the plates are wide ($w \gg d$), the magnetic field between the plates is uniform. By Ampere's Law, the magnetic flux density B in the dielectric gap is:

$$B = \frac{\mu_0 \mu_r I}{w} \quad (3.17)$$

The total magnetic flux Φ linking the loop formed by the two plates over a length l and a dielectric separation d is the product of the flux density and the cross-sectional area of the gap ($A = l \cdot d$):

$$\Phi = B \cdot A = \left(\frac{\mu_0 \mu_r I}{w} \right) (l \cdot d) \quad (3.18)$$

Since inductance is defined as flux linkage per unit current ($L = \Phi/I$), the equation simplifies to the standard parallel-plate transmission line inductance:

$$L_{loop} = \frac{\mu_0 \mu_r \cdot l \cdot d}{w} \quad (3.19)$$

Conclusion: This proof confirms that loop inductance is governed entirely by the geometric ratio of l/w and the dielectric gap d . By utilizing an ultra-thin dielectric (e.g., 2 mil or 0.050 mm Kapton) and maximizing width w , the fixture's parasitic inductance is mathematically forced into the sub-nanohenry regime.

3.3.3 Equivalent Inductance of Coupled Terminals (L_{eq})

In multi-terminal capacitors, same-polarity pins run in parallel and share a magnetic boundary. When two parallel conductors are magnetically coupled, their voltage-current relationships are:

$$v = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \quad (3.20)$$

$$v = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt} \quad (3.21)$$

Because the pins are connected in parallel, the total current is $i = i_1 + i_2$, and the voltage v across both branches is equal. Solving this system for the equivalent inductance ($L_{eq} = v / \frac{di}{dt}$) yields the general coupled parallel inductance formula:

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \quad (3.22)$$

For symmetrical capacitor terminals, $L_1 = L_2 = L_{pin}$. Substituting this into the general formula allows us to factor and simplify:

$$L_{eq} = \frac{L_{pin}^2 - M^2}{2L_{pin} - 2M} = \frac{(L_{pin} - M)(L_{pin} + M)}{2(L_{pin} - M)} \quad (3.23)$$

$$L_{eq} = \frac{L_{pin} + M}{2} \quad (3.24)$$

Impact of Mutual Coupling:

Ideal uncoupled state ($M \rightarrow 0$): $L_{eq} = L_{pin}/2$. This is the naive assumption, yielding a perfect 50% reduction in inductance.

Tightly coupled state ($M \rightarrow L_{pin}$): $L_{eq} \rightarrow L_{pin}$. Paralleling the pins provides negligible reduction because shared magnetic flux dominates.

To correct this, the H-bridge plates act as a magnetic mirror. The opposing differential currents in the laminated plates generate a strong opposing magnetic field that induces eddy currents in the surrounding conductive environment. This produces an opposing flux (Φ_{ind}) that cancels the mutual flux linking the pins:

$$\Phi_{M, effective} = \Phi_M - \Phi_{ind} \Rightarrow M_{effective} \rightarrow 0 \quad (3.25)$$

By forcing $M \rightarrow 0$, the laminated fixture restores the theoretical factor-of-two inductance reduction, preventing the systematic overestimation of DUT ESL that occurs in standard open fixtures.

3.3.4 Vector Orthogonality and Flux Decoupling

To prevent the H-bridge plates from also suppressing the pins' own self-inductance (L_{pin}) via the same eddy-current mechanism—which would falsely lower the measured ESL—the fixture enforces a 90° spatial rotation between the current paths. We can define the current density vectors mathematically. The pin current flows vertically along the z-axis, while the plate current flows horizontally in the xy-plane:

$$\mathbf{J}_{pin} = J_z \hat{k} \quad (3.26)$$

$$\mathbf{J}_{plate} = J_x \hat{i} + J_y \hat{j} \quad (3.27)$$

The dot product of these two current densities is strictly zero:

$$\mathbf{J}_{pin} \cdot \mathbf{J}_{plate} = 0$$

Because the currents are orthogonal, the magnetic fields they generate are also orthogonal to each other's primary flux linkage loops. The magnetic flux circulating around the pins is parallel to the flat surfaces of the busbar plates. Consequently, the pins' self-generated flux does not enter the inter-plate volume, preventing orthogonal flux compression. This orthogonality guarantees that only inter-pin mutual coupling (M) is cancelled, fully preserving the self-inductance contribution of the capacitor under test.

CHAPTER 4

EXPERIMENTAL SETUP

4.1 ESL Setup Circuit

The test circuit including the loop resistance and loop inductance, a switch is shown in Figure 4.1.

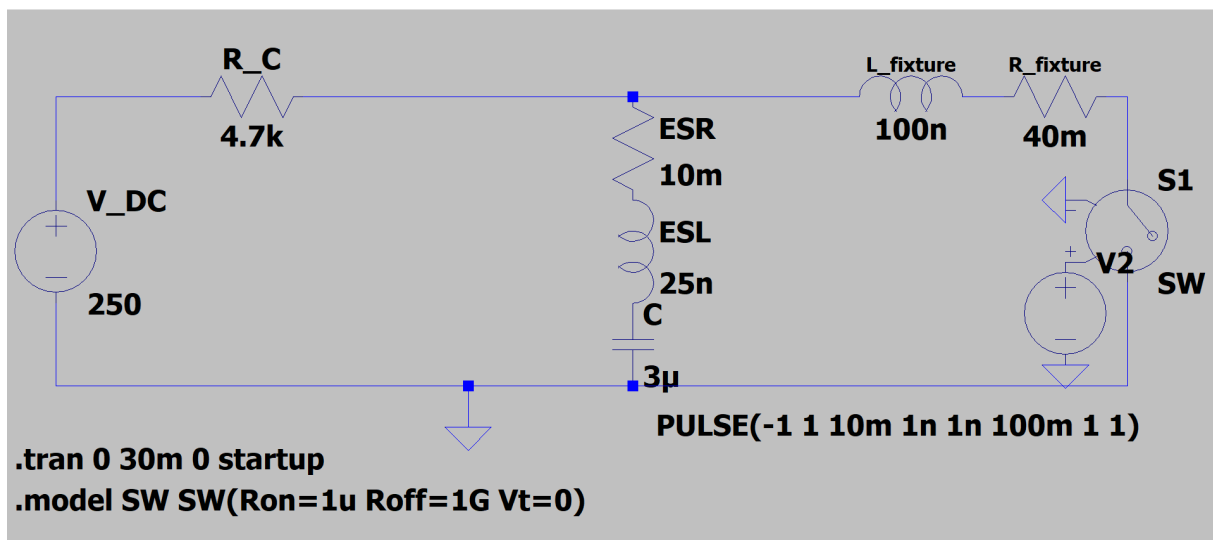


Figure 4.1: [Equivalent test circuit]

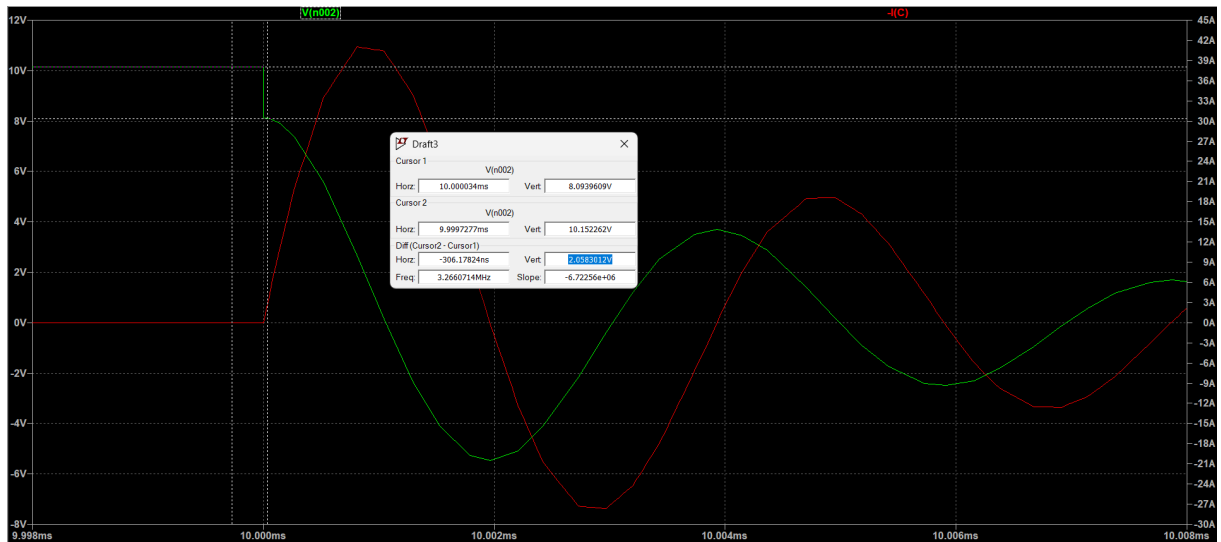


Figure 4.2: [Equivalent test circuit voltage waveform]

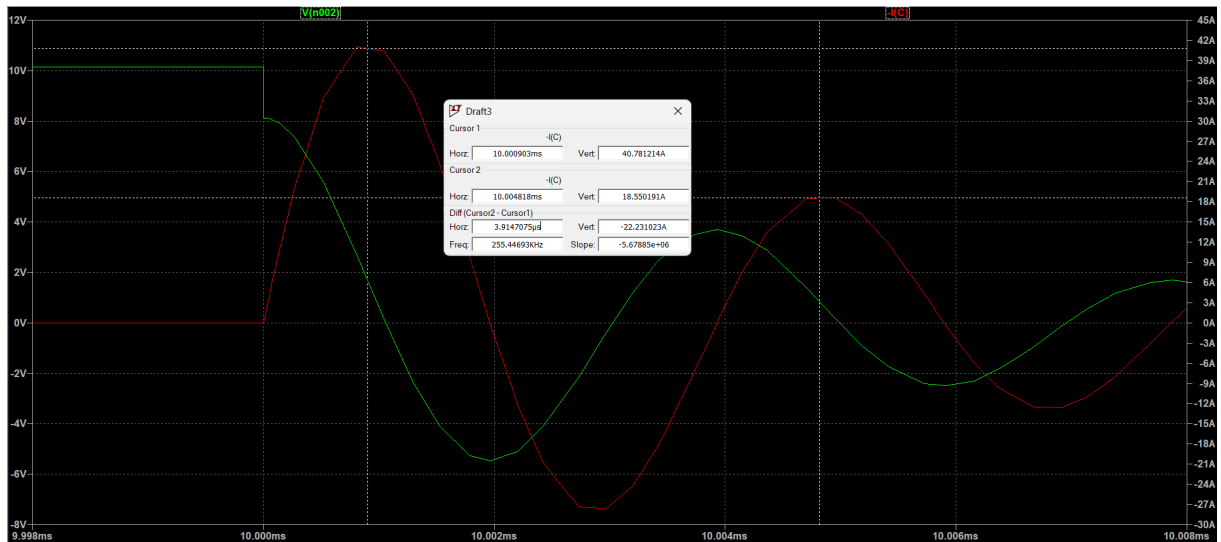


Figure 4.3: [Equivalent test circuit current waveform]

Based on these values:

$$V_1 = 2.0583 \text{ V}$$

$$V_C = 10.1522 \text{ V}$$

$$I_{p1} = 40.8 \text{ A}$$

$$I_{p2} = 18.5 \text{ A}$$

$$T = 3.92 \text{ us}$$

$$C = 3 \text{ uF}$$

$$\text{Thus, } L_C = 25.89 \text{ nH}$$

4.2 ESL Circuit with Chosen SiC switch

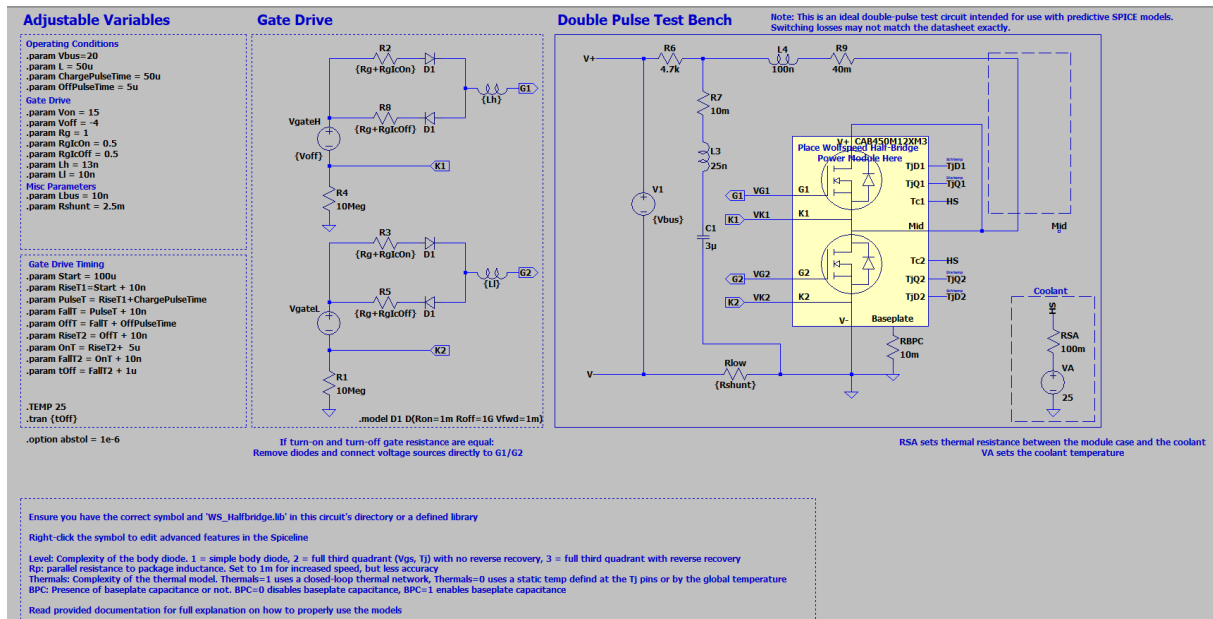


Figure 4.4: [Realistic simulation circuit of SiC switch for capacitor charge-discharge]

Adjustable Variables

Operating Conditions

```
.param Vbus=20
.param L = 50u
.param ChargePulseTime = 50u
.param OffPulseTime = 5u
```

Gate Drive

```
.param Von = 15
.param Voff = -4
.param Rg = 1
.param RgIcOn = 0.5
.param RgIcOff = 0.5
.param Lh = 13n
.param Ll = 10n
```

Misc Parameters

```
.param Lbus = 10n
.param Rshunt = 2.5m
```

Gate Drive Timing

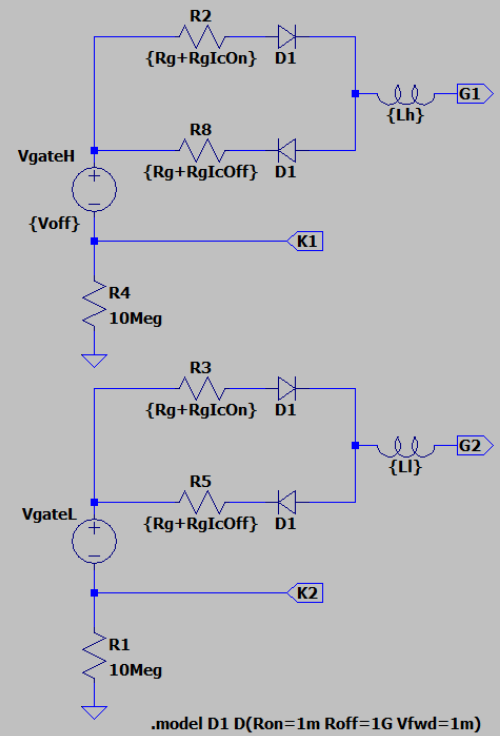
```
.param Start = 100u
.param RiseT1=Start + 10n
.param PulseT = RiseT1+ChargePulseTime
.param FallT = PulseT + 10n
.param OffT = FallT + OffPulseTime
.param RiseT2 = OffT + 10n
.param OnT = RiseT2+ 5u
.param FallT2 = OnT + 10n
.param tOff = FallT2 + 1u
```

```
.TEMP 25
```

```
.tran {tOff}
```

```
.option abstol = 1e-6
```

Gate Drive



If turn-on and turn-off gate resistance are equal:
Remove diodes and connect voltage sources directly to G1/G2

Figure 4.6: [MOSFET Gate Drive]

Figure 4.5: [Adjustable variables list]

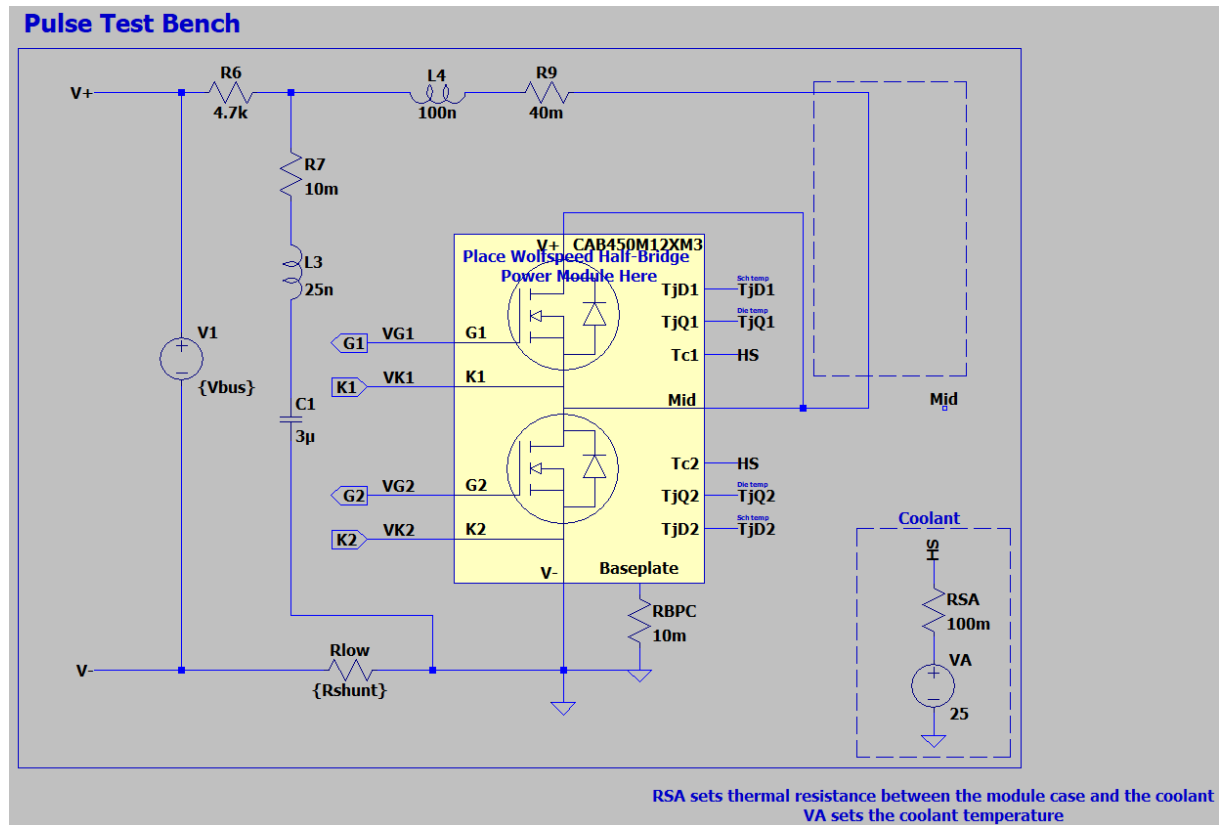


Figure 4.7: [Closer look at the circuit]

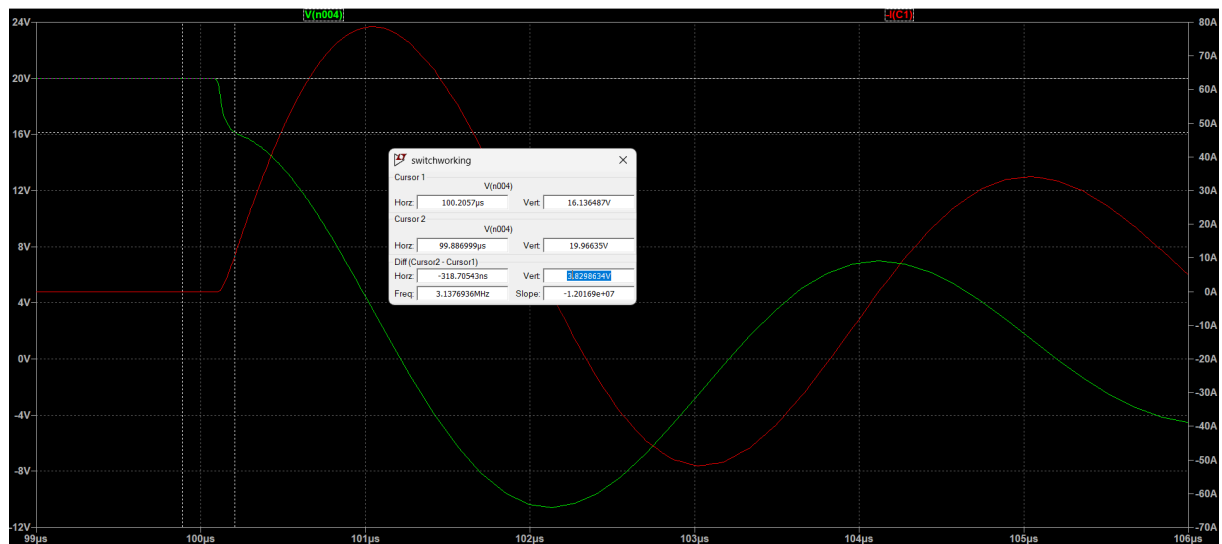


Figure 4.8: [SiC Voltage Waveform]

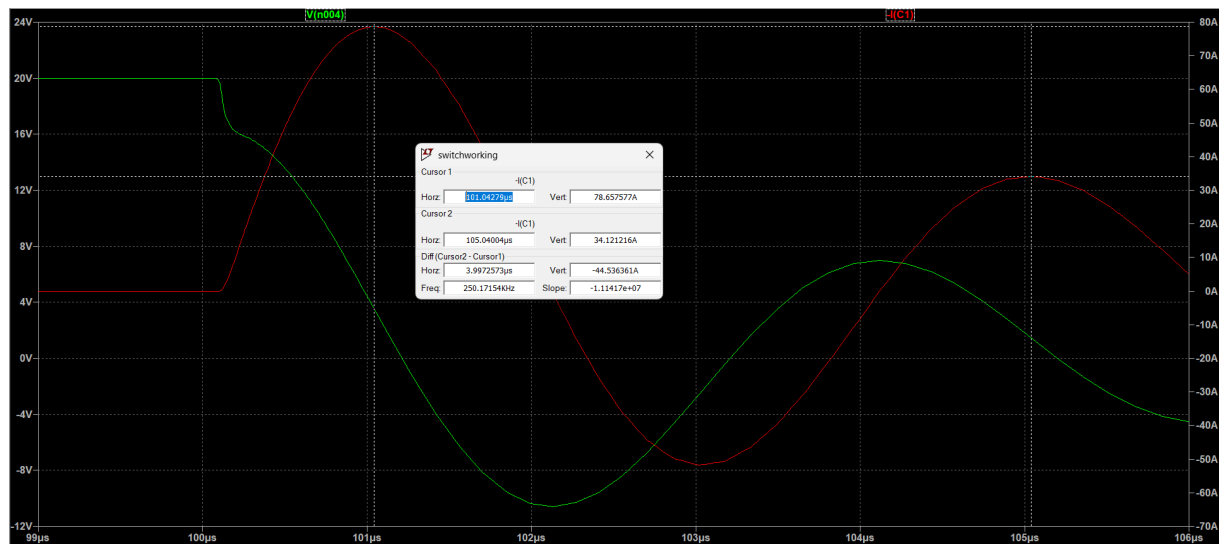


Figure 4.9: [SiC Current Waveform]

Based on these values:

$$V_1 = 3.8 \text{ V}$$

$$V_C = 19.96 \text{ V}$$

$$I_{p1} = 78.65 \text{ A}$$

$$I_{p2} = 34.1 \text{ A}$$

$$T = 4 \text{ } \mu s$$

$$C = 3 \text{ } \mu F$$

$$\text{Thus, } L_C = 25.27 \text{ nH}$$

4.3 Piano Setup Practical images

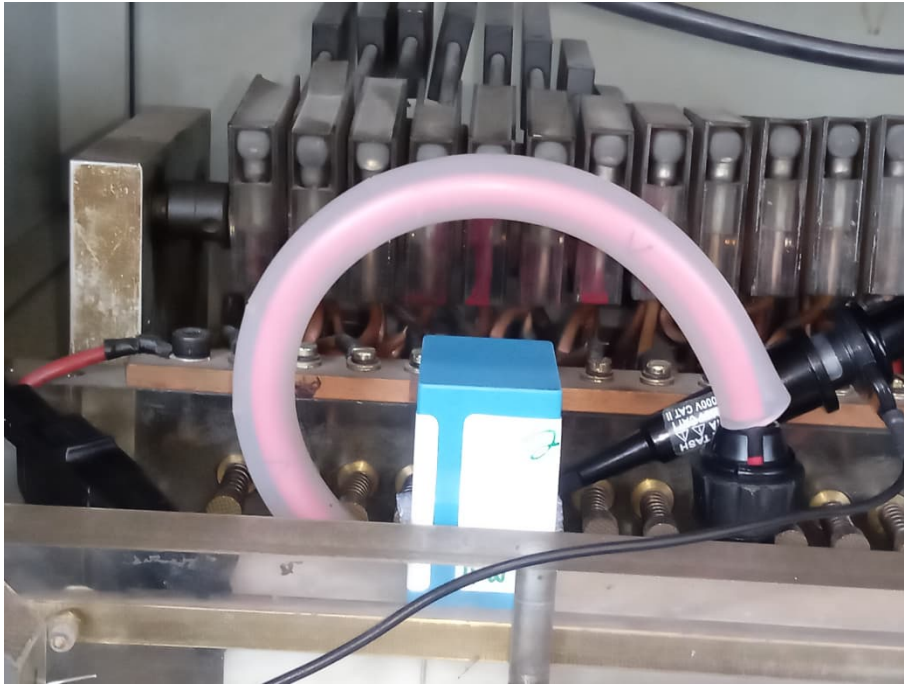


Figure 4.10: [Piano setup front view]



Figure 4.11: [Piano setup top view]

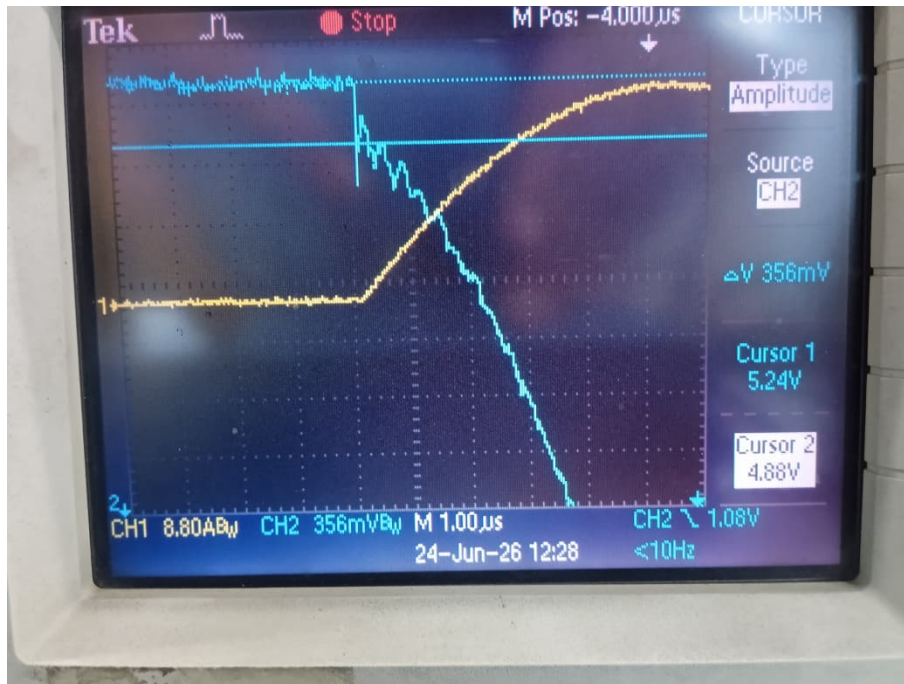


Figure 4.12: [Resolution problem faced when $L_C \ll L_{\text{fixture}}$]

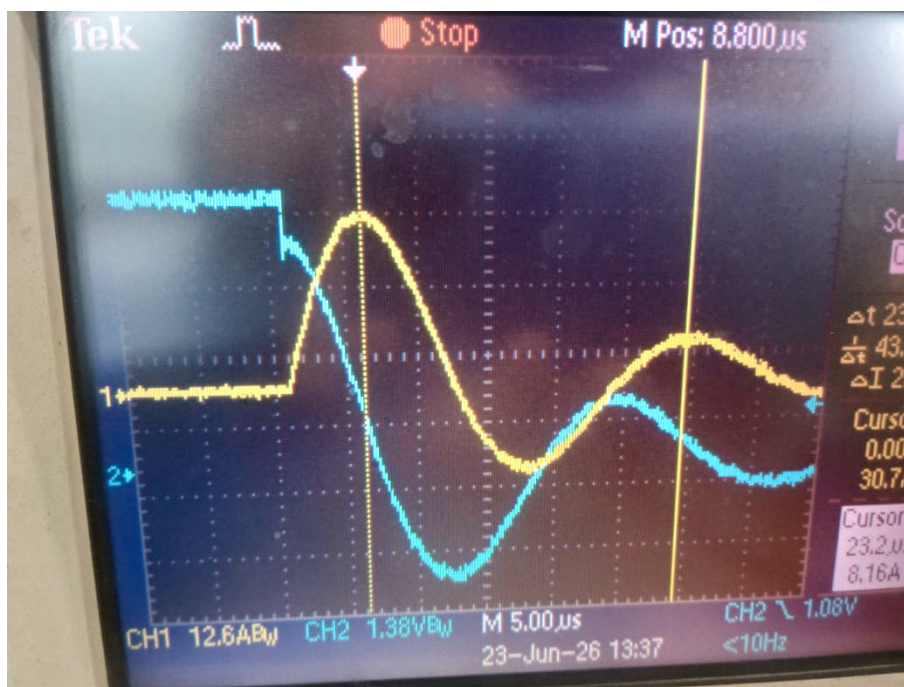


Figure 4.13: [How a clean discharge waveform should be for accurate readings]

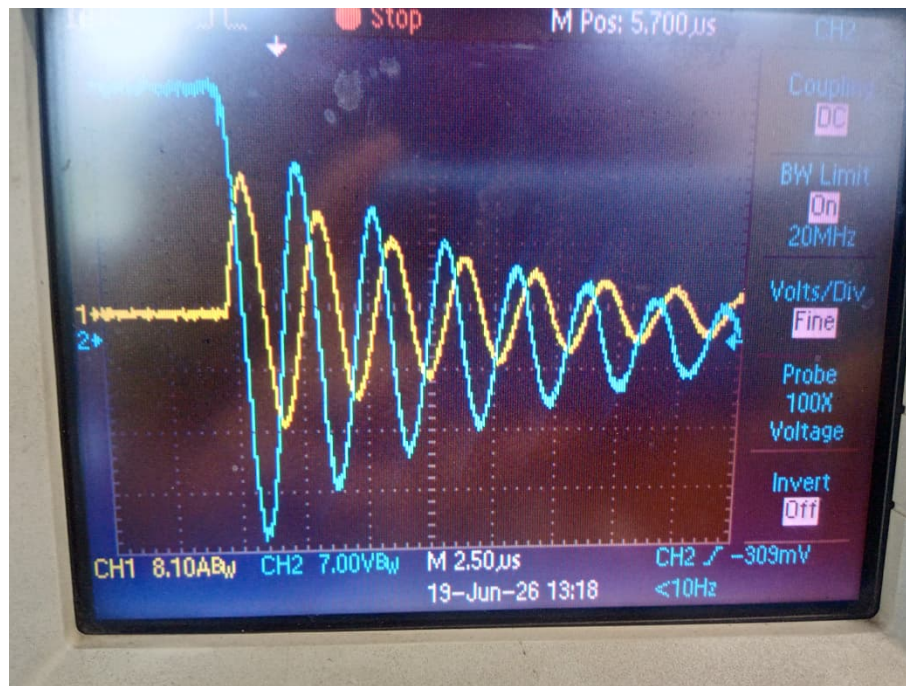


Figure 4.14: [Initial damping waveform]

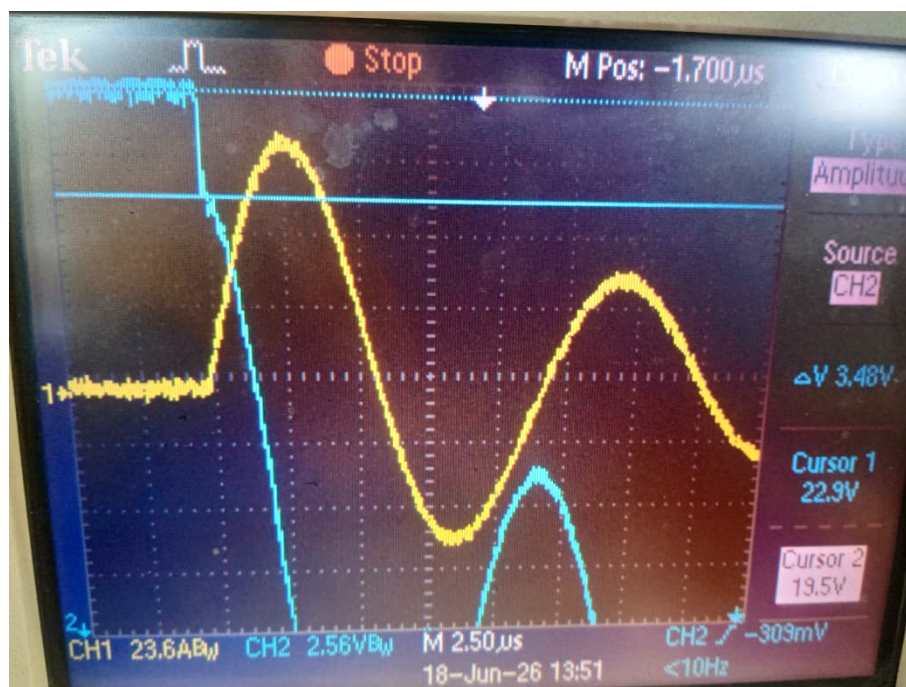


Figure 4.15: [zoomed in damping waveform]

4.4 Proposed test setup with copper bridge

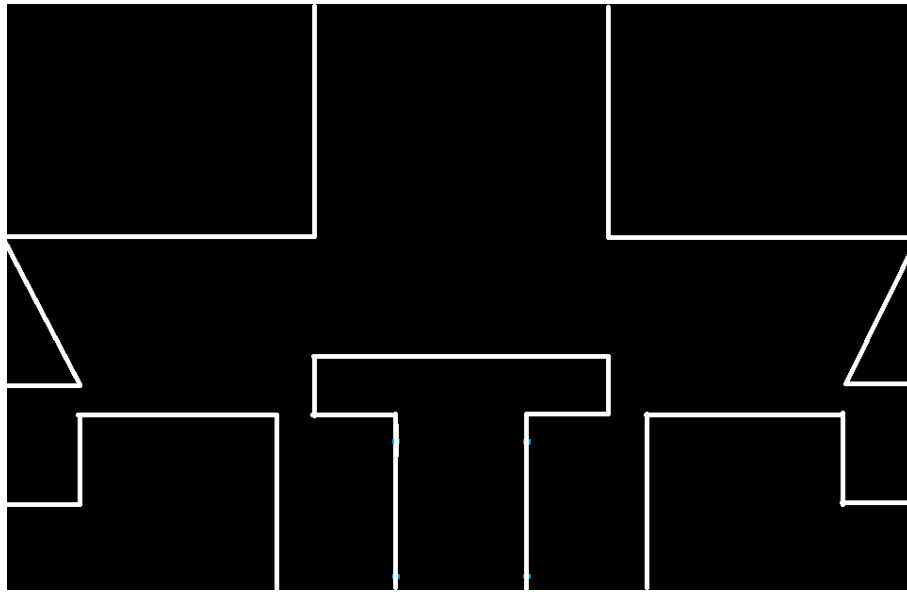


Figure 4.16: [A sample schematic for Copper H Bridge]

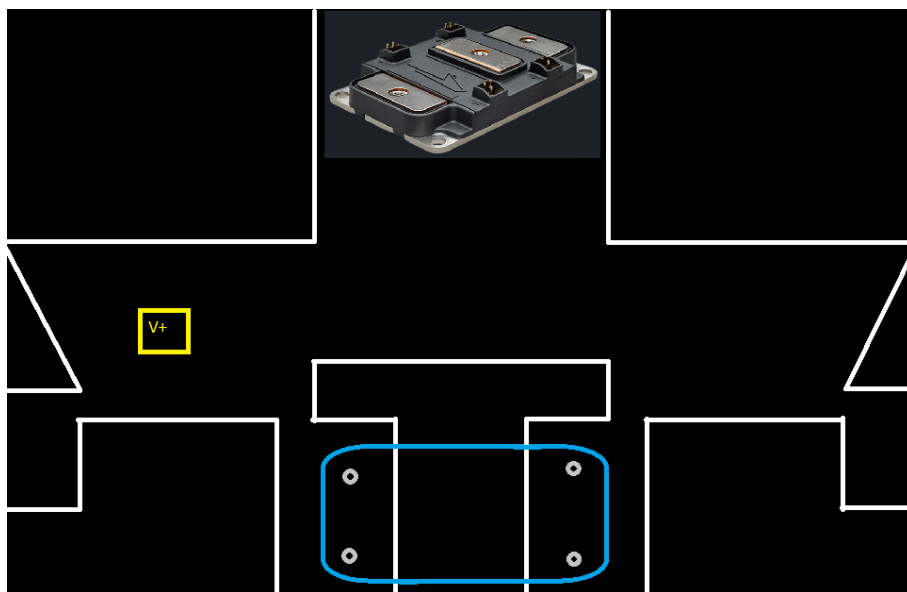


Figure 4.17: [Copper bridge fixture schematic]

The fixture consists of 2 parallel copper plates separated by a thin dielectric. The positive terminal pair of the capacitor is connected to the top plate and negative terminal pair to the bottom plate using two through holes. The DC power supply is given to the top plate thus making it the positive plate and bottom the reference ground. The switch is connected between the two plates which shorts the circuit to initiate capacitor discharge.

CHAPTER 5

RESULTS AND ANALYSIS

5.1 B32922 0.33 uF 2 terminal capacitor

5.1.1 Sample 1

Table 5.1: Readings for B32922 sample 1

Parameter	set 1	set 2	set 3	set 4	set 5
V_{op}	242	259	290	273	245
V_c	252	265	299	280	252
V_1	11.8	13.4	13.9	13.6	12.2
V_2	240	252	286	266	240
I_1 (A)	174	186	215	197	179
I_2 (A)	79.4	85.2	104	94.5	80.7
T (μ s)	2.52	2.5	2.52	2.5	2.52
C (μ F)	0.31483	0.31483	0.31483	0.31483	0.31483
a (10^6)	0.311332325	0.312298306	0.288193499	0.293841755	0.316130066
ω_r (M rad/s)	2.493327503	2.513274123	2.493327503	2.513274123	2.493327503
ω_0^2	6.313609853	6.414077049	6.299737529	6.402889794	6.316620255
ω_0 (M rad/s)	2.512689765	2.532602821	2.509927794	2.530393209	2.513288733
$1/L$	1.98771379	2.019343877	1.983346366	2.015821794	1.988661555
L (nH)	503.090538	495.2103558	504.1983674	496.0755971	502.8507729
L_c (nH)	23.55741408	25.04082554	23.4393221	24.09510043	24.34436282

5.1.2 Sample 2

Table 5.2: Readings for B32922 sample 2

Parameter	set 1	set 2	set 3
V_{op}	22	21	30
V_c	26.8	25.7	30
V_1	1.41	1.3	1.51
V_2	25.4	24.3	28.5
I_1 (A)	21.4	16.2	36.45
I_2 (A)	14.3	12	27
T (μ s)	2.28	2.4	2.4
C (μ F)	0.3138283	0.3138283	0.3138283
a (10^6)	0.17681213	0.125043664	0.125043664
ω_r (M rad/s)	2.755783029	2.617993878	2.617993878
ω_0^2	7.625602635	6.869527863	6.869527863
ω_0 (M rad/s)	2.761449372	2.620978417	2.620978417
$1/L$	2.393129911	2.155852251	2.155852251
L (nH)	417.8628144	463.8536799	463.8536799
L_c (nH)	21.98457344	23.46341571	23.34730189

Table 5.3: Statistics for B32922

average	23.6590395
median	23.5104149
skew	-0.456908075
variance	0.795291269
Standard Deviation	0.891791045

The theoretical value for a 2 terminal capacitor with $d=0.75$ mm and $l=15$ mm is (22.44 - 24.44)nH

5.2 R&D Samples 30uF: 2 terminal and 4 terminal

5.2.1 2 terminal

Table 5.4: Readings for R&D sample 1

Parameter	set 1	set 2	set 3
V_{op}	10	20	28
V_c	4.05	7.45	10.2
V_1	0.65	1.23	1.63
V_2	3.39	6.22	8.57
I_1 (A)	24.4	43.6	58.9
I_2 (A)	7.2	13.4	18.4
T (μ s)	24.4	24	24
C (μ F)	28.54253	28.54253	28.54253
a (10^6)	0.050020612	0.049158468	0.0484788
ω_r (M rad/s)	0.25707595	0.261799388	0.261799388
ω_0^2	0.068812223	0.070955474	0.070889114
ω_0 (M rad/s)	0.26232084	0.266374688	0.266250096
$1/L$	1.964074935	2.025248758	2.02335465
L (nH)	509.1455433	493.7665045	494.2287305
R (m Ω)	50.93554311	48.54561003	47.91923192
L_c (nH)	81.71471683	81.52118128	78.97968929

Table 5.5: Readings for R&D sample 2

Parameter	set 1	set 2	set 3
V_{op}	9	20	30
V_c	4.44	7.53	11.1
V_1	0.8	1.34	1.99
V_2	3.64	6.18	9
I_1 (A)	25.9	44.4	64.2
I_2 (A)	8.35	15.1	22
T (μ s)	24	24	24
C (μ F)	28.47062	28.47062	28.47062
a (10^6)	0.047165925	0.044939394	0.044623395
ω_r (M rad/s)	0.261799388	0.261799388	0.261799388
ω_0^2	0.070763544	0.070558469	0.070530167
ω_0 (M rad/s)	0.26601418	0.265628441	0.26575162
$1/L$	2.014681968	2.008843346	2.008037578
L (nH)	496.352566	497.798896	497.9986485
R (m Ω)	46.82220358	44.74156123	44.4447807
L_c (nH)	89.43355975	88.58572651	89.28083878

Table 5.6: Statistics

average	84.91928541
median	85.15022167
skew	-0.167120107
variance	21.98795689
Standard Deviation	4.689131785

The theoretical value for a 2 terminal capacitor with d=1.2mm and l=42mm is (68.46 - 78.46)nH

5.2.2 4 terminal

Table 5.7: Readings for R&D sample 1

Parameter	set 1	set 2	set 3
V_{op}	12	21	28
V_c	5.3	8.44	10.8
V_1	0.654	1.02	1.34
V_2	4.57	7.42	9.43
I_1 (A)	30.9	49.9	64.8
I_2 (A)	8.74	14.1	19
T (μ s)	23.2	23.4	23.2
C (μ F)	28.70805	28.70805	28.70805
a (10^6)	0.054433054	0.054010558	0.052882218
ω_r (M rad/s)	0.270826953	0.268512193	0.270826953
ω_0^2	0.076310196	0.075015938	0.076143767
ω_0 (M rad/s)	0.276243001	0.273890376	0.275941601
$1/L$	2.190716915	2.153561297	2.185939081
L (nH)	456.4715748	464.3471264	457.4692904
R (m Ω)	49.6942834	50.15929474	48.38398116
L_c (nH)	56.32686979	56.11778068	56.76007862

Table 5.8: Readings for R&D sample 2

Parameter	set 1	set 2	set 3
V_{op}	12	20	30
V_c	4.89	7.9	11.6
V_1	0.529	0.874	1.29
V_2	4.36	7.03	10.3
I_1 (A)	28.7	48.4	70.5
I_2 (A)	7.73	14.4	21.4
T (μ s)	23.4	23.4	23.4
C (μ F)	28.59019	28.59019	28.59019
a (10^6)	0.056059365	0.051806514	0.050949683
ω_r (M rad/s)	0.268512193	0.268512193	0.268512193
ω_0^2	0.07524145	0.074782712	0.074694668
ω_0 (M rad/s)	0.27430175	0.27346428	0.273303252
$1/L$	2.151167351	2.138051958	2.135534745
L (nH)	464.8638793	467.7154811	468.2667901
R (m Ω)	52.11994784	48.46141704	47.71608933
L_c (nH)	50.28895545	51.74472538	52.07449648

Table 5.9: Statistics

average	53.8854844
median	54.09613858
skew	-0.170239937
variance	8.000869632
Standard Deviation	2.828580851

The theoretical value for a 4 terminal capacitor with $d=1.2\text{mm}$ and $l=42\text{mm}$ is $(35.23 - 37.23)\text{nH}$. We observe a higher reading due to noncompensated mutual inductance.

5.2.3 4 terminal with parallel copper plate

Table 5.10: Capacitor Discharge readings with copper plate

Parameter	set 1	set 2	set 3
V_{op}	12	21	30
V_c	5.24	8.18	11.5
V_1	0.384	0.6	0.896
V_2	4.86	7.58	10.6
I_1 (A)	30.3	48.8	68.9
I_2 (A)	9.15	15.5	21.6
T (μ s)	23.6	23.6	23.6
C (μ F)	28.70805	28.70805	28.70805
a (10^6)	0.050737061	0.048597079	0.049151002
ω_r (M rad/s)	0.266236666	0.266236666	0.266236666
ω_0^2	0.073456211	0.073243638	0.073297783
ω_0 (M rad/s)	0.271028064	0.270635619	0.270735633
$1/L$	2.108784591	2.102682026	2.104236421
L (nH)	474.2068034	475.5830827	475.2317705
R (m Ω)	48.11971899	46.22389712	46.71623526
L_c (nH)	34.75103292	34.88384469	37.0267536

The theoretical value for a 4 terminal capacitor with $d=1.2\text{mm}$ and $l=42\text{mm}$ is (35.23 - 37.23)nH. We observe a similar reading, but there is a chance of overcompensation of mutual inductance and reduced values due to parallel geometry.

CHAPTER 6

CONCLUSION

This project successfully established a unified setup for measuring the Equivalent Series Inductance (ESL) of film capacitors by integrating the low-voltage surge discharge method with Silicon Carbide (SiC) MOSFET switching technology. The proposed system offers a robust, high-volume, and time-domain alternative to traditional impedance analyzers, simplifying instrumentation while maintaining high precision for nanohenry-scale ESL measurements in industrial R&D environments.

The principal findings and contributions of this work are:

- **Validation of the Surge Discharge Method:** The project standardized a calibration-free technique that extracts ESL by analyzing the underdamped ringing current waveform of a discharging capacitor. This approach achieved a combined measurement uncertainty of ± 0.151 nH, outperforming traditional Auto-Balanced Bridge (ABB) and TDR methods. For the B32922 0.33 μ F 2-terminal capacitor, the experimental mean ESL was 23.66 nH ($\sigma = 0.89$ nH), which closely aligns with the theoretical expectation of 22.44–24.44 nH.
- **Integration of SiC MOSFETs as a Sustainable Switch:** To comply with the Minamata Convention’s phase-out of toxic mercury-wetted relays, the Wolfspeed CAB450M12XM3 SiC MOSFET was characterized as a sustainable, RoHS-compliant solid-state alternative. The SiC device provides a high blocking voltage capability (> 1.2 kV), nanosecond transition speeds (2–10 ns), and a high current rise rate of 38.7 kA/ μ s. SPICE simulations confirmed that the SiC switch introduces minimal measurement error, yielding an inductance of 25.27 nH against an ideal 25.89 nH (a 2.4% deviation).
- **Development of a Laminated Copper H-Bridge Fixture:** Initial ESL measurements for 4-terminal capacitors were significantly overestimated (e.g., measuring 53.88–53.89 nH against a theoretical 35.23–37.23 nH) due to uncompensated mutual inductance between parallel pins. To mitigate this magnetic coupling, a high-precision laminated copper H-bridge fixture was proposed. By utilizing orthogonal flux decoupling, this fixture effectively

cancels the mutual inductance, bringing the measured ESL within the theoretical target range.

Future Work

Future metrology improvements should focus on the following areas:

- Physical fabrication and experimental hardware validation of the proposed laminated copper H-bridge fixture and the complete SiC-based switching circuit.
- Automated waveform processing and parameter extraction to eliminate manual cursor errors and improve measurement repeatability for routine laboratory testing.
- Extending the methodology to reliably characterize capacitors with an ESL below 10 nH, where signal-to-noise ratio (SNR) constraints and shrinking V_1/V_C ratios heavily dominate measurement uncertainty.
- Evaluating the system across a broader range of capacitor technologies and compensating for residual fixture parasitics through advanced calibration techniques.

Overall, the integration of advanced wide-bandgap switching with optimized fixtures ensures a compact and environmentally compliant system that can confidently maintain the strict inductance limits (10 nH to 50 nH) required by modern high-power systems and emerge as a reliable lab testing method.